

INFOSCI201 Final Project Proposal

1. Project Concept (250–500 words)

This project is a two-player interactive racing game that integrates a random luck-based mechanism with strategic decision-making. The game is inspired by the Mongolian shagai tradition, where four distinct shapes determine outcomes and two horses race to the finish line.

In this game, an Arduino Uno acts as the physical controller. Each roll generates four shapes: ♥, ▲, ■, and ●. Only ♥ (hearts) move the player's horse forward, while ▲ (triangles), ■ (squares), and ● (circles) function as an in-game currency that players can combine to activate special powers. Hearts are never stored as currency; they are used immediately for movement. The other three shapes accumulate separately for each player and persist across turns.

Unity displays two parallel roads made of square tiles, one for each player. Horses advance along their respective tracks based on the number of ♥ shapes rolled. Before rolling, players may spend their collected shapes to activate skills. These skills are intentionally expensive, difficult to acquire, and do not guarantee movement. Therefore, it doesn't give an unfair advantage to any player. Instead, they create tactical opportunities, psychological tension, and dramatic moments. The Left and Right buttons on the Arduino are used to navigate the power menu and position certain abilities on the road, while the Confirm button is used to purchase and activate them. A separate Roll button triggers the next set of four shapes. All currency and game information is displayed visually at the top of the Unity screen.

The buzzer provides audio feedback for rolls, special events, and the end. Two LED lights with different colors on two sides indicate which player's turn it is. The tutorial button provides access to basic configuration options and a simple explanation of the rules, controls, and skill system.

The game heavily relies on Unity, and the goal is to make it fully controllable using only the Arduino board. This project has a theme of horse racing, but it is an enriched and competitive version of a traditional Mongolian game.

2. Sensors Used

- Push buttons x4 (Rolling, confirmation, navigation)

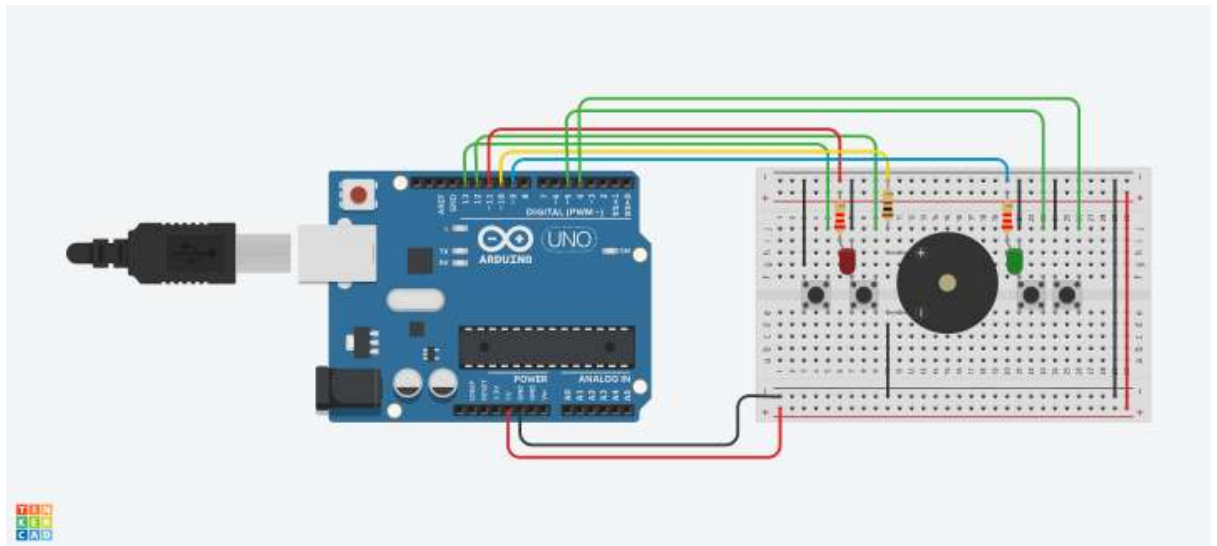
3. Other Components

- Piezo buzzer x1
- LED lights x2
- USB connection for serial communication

4. Additional Materials

- None

5. Circuit Sketch



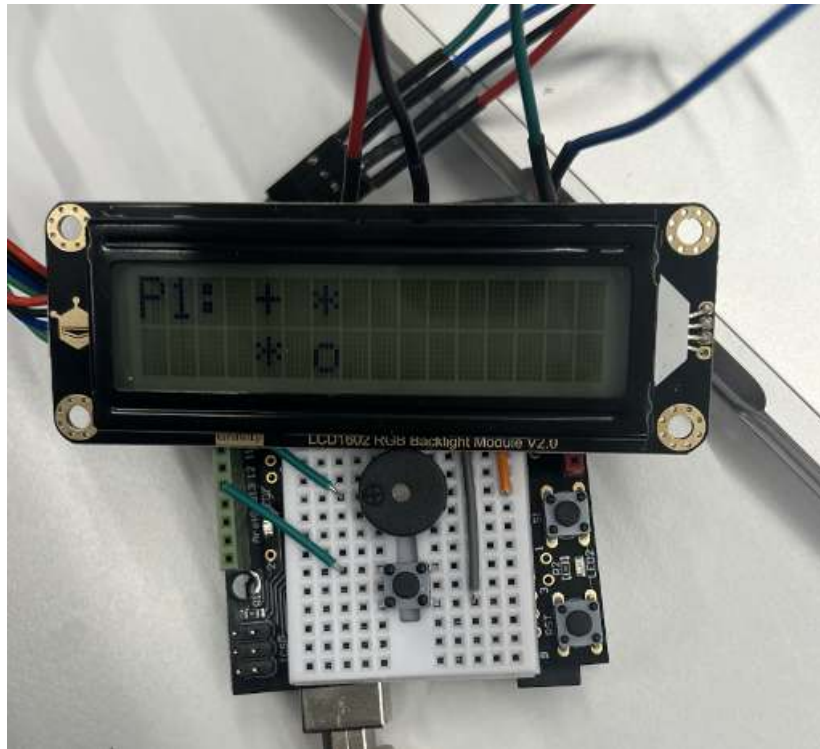
6. Project process and iterations

1. Initial Concept & System Planning

- Defined the core idea: a two-player racing game inspired by Mongolian shagai.
- Established the four shapes: ♥, ▲, ■, ●.
- Decided that ♥ hearts move the horse, while ▲ triangles, ■ squares, and ● circles act as currency.
- Designed the skill system: Mud Trap, Whip Boost, Foresight.
- Planned full Arduino-only control: roll, confirm, left, right.
- Designed UI elements on Unity: roll display, currency bar, power menu, foresight UI, and GameOver panel.

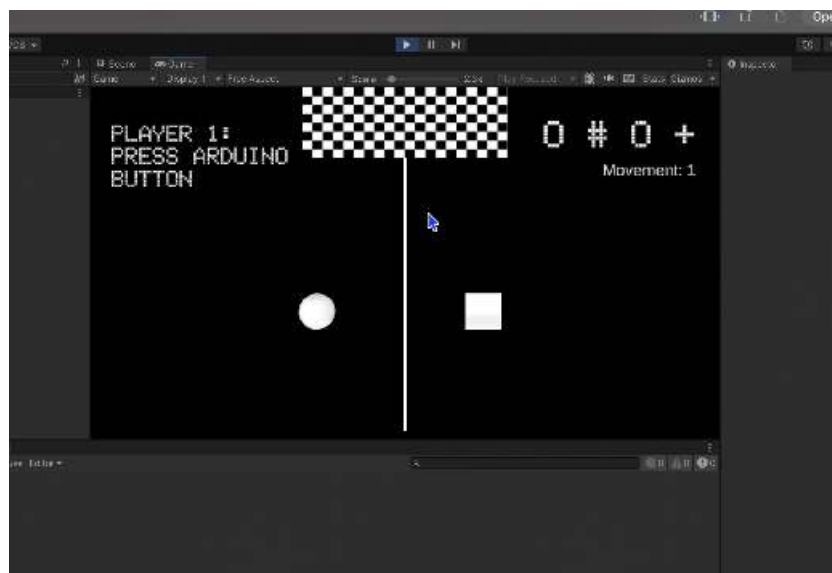
2. Prototype Development

- Used Unity to build a 2D prototype of the game without any power or currency system
- Built a simple Arduino controller



3. Arduino to Unity Communication

- Implemented serial communication for shape values.
- Created a 4-roll buffer to collect all shapes before processing.
- Added LED turn indicators and buzzer feedback.



4. Track & Horse Movement System

- Built tile-based lanes for both players.
- Implemented horse movement based on the number of ♥ hearts.
- Added finish line detection and victory handling.
- Then tested the basic functions of the game.



5. UI System & Visual Feedback

- Improved the UI by adding 3D elements from imported assets.
- Adjusted the fonts and size of the text
- Added a Tutorial page

Skill system rules:



1. These skills use ▲ (triangles), ■ (squares), and ● (circles) as currency.
2. Players can activate skills only before rolling.
 - Mud Trap (■2 + ▲1)
 - Whip Boost (■4 + ●2)
 - Foresight (●3 + ▲2)

Some major problems I faced:

- Implementing animation on horses. Even though the 3D asset package came with animations, it was not working as it should.
- Generating and addressing each tile.
- Showing the Game Over screen once the a player wins.

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INSTRUCTIONS

HOW TO PLAY

- Press R to roll 4 symbols.

- ♥ = Movement. Each heart moves your horse forward 1 tile.
- ■, ▲, ● = Currency. Spend these on powers.

POWERS

- Mud Trap: Place a trap on the opponent's lane. They move back 2 when stepping on it.
- Whip Boost: Your next movement is doubled.
- Foresight: Guess how many ♥ you will roll. If correct, take another turn.

CONTROLS

- R = Roll
- C = Open/Use Powers
- ← → = Navigate menus or place traps

GOAL

Reach the final tile before your opponent.